

INTERMEDIATE QUANT 1

Directions for question 1: Select the correct alternative from the given choices.

Rajesh and Sumit left Mumbai for Pune in their respective cars at 10:00 am. Both reached Pune at 3:00 pm but Rajesh drove for three times as long as Sumit rested on his trip, and Sumit drove for five times as long as Rajesh rested on his trip. What is the ratio of Rajesh's and Sumit's speeds?

- ☐ a) 5 : 6
- ☐ b) 1 : 2
- ☐ c) 4 : 5
- ☐ d) None of these

Solution:

The data is tabulated below.

	Drove	Rested
Rajesh	3x	y
Sumit	5y	x

$$3x + y = 5y + x \Rightarrow x = 2y$$

$$\therefore 3x = 6y.$$

Rajesh drove for time 6y and Sumit for 5y to cover the same distance.

$\therefore$ The ratio of their speeds is 5 : 6.

Choice (A)

Directions for question 2: Type in your answer in the input box provided in the question.

If $f(x) = ax + b$, where a and b are constants and

$$f[f(f(x))] = 8x + 35$$

, then $2a + 3b =$

Your Answer: 25 ✖

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Solution:

$$\begin{aligned} f(x) &= ax + b \\ f^2(x) &= f(f(x)) = a^2x + ab + b \\ f^3(x) &= a^3x + a^2b + ab + b \\ &= a^3x + (a^2 + a + 1)b = 8x + 35 \text{ for all } x \\ a &= 2, b = 5 \text{ and } 2a + 3b = 4 + 15 = 19 \end{aligned}$$

Ans : (19)

Directions for questions 3 and 4: Select the correct alternative from the given choices.

How many integral solutions are there for the equation $32x - 112y = 1000$?

- ☐ a) 2
- ☐ b) 10
- ☐ c) infinity
- ☐ d) None of these

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Solution:

$$32x - 112y = 1000$$

The LHS is a multiple of 16 while the RHS is not.

There are no solutions.

Choice (D)

Directions for questions 3 and 4: Select the correct alternative from the given choices.

The number of integer values of x that satisfy the inequation

$$\left| \frac{1}{2x-9} \right|$$

>

$$\frac{3}{7}$$

$$\left(x \neq \frac{9}{2} \right)$$

is _____.

- ☐ a) 1
- ☐ b) 2
- ☐ c) 3
- ☐ d) 4

Solution:

$$\left| \frac{1}{2x-9} \right| > \frac{3}{7}$$

$$\Rightarrow |2x-9| < \frac{7}{3}$$

$$\Rightarrow -\frac{7}{3} < 2x-9 < \frac{7}{3}$$

$$\Rightarrow -\frac{7}{3} + 9 < 2x < \frac{7}{3} + 9$$

$$\frac{20}{3 \times 2} < x < \frac{34}{3 \times 2}$$

$$\frac{10}{3} < x < \frac{17}{3}$$

$$3.3 < x < 5.6$$

$\therefore$ 4 and 5 are integer values of x that satisfy the given inequation.

Choice (B)

Directions for question 5: Type in your answer in the input box provided below the question.

The minimum distance between the circles

$x^2 + y^2 + 4x + 16y + k = 0$ and $x^2 + y^2 - 6x - 8y + 16 = 0$ is 8. Find the value of k.

Your Answer: Unattempted

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Solution:

The first circle is $(x+2)^2 + (y+8)^2 = 68 - k$.

The second circle is $(x-3)^2 + (y-4)^2 = 9$.

The first circle is centered at $(-2, -8)$ and has a radius, r_1 of $\sqrt{68 - k}$.

The second circle is centered at $(3, 4)$ and has a radius r_2 of 3.

Distance between $(-2, -8)$ and $(3, 4)$ is $D = 13$. The distance(d) between any point on the first circle and any point on the second circle is minimum (or maximum) when the points lie on the line joining the centres of the two circles.

$\therefore d_{\min} = D - (r_1 + r_2)$.

$= 13 - (\sqrt{68 - k} + 3) = 8$ (given)

$\Rightarrow k = 64$

Ans : (64)

Directions for questions 6 to 10: Select the correct alternative from the given choices.

R takes three times as much time as P takes to complete a certain work and twice as much time as Q takes to complete the same work. If P, Q and R can together complete the work in 10 days, how much time (in days) would each of them individually take to complete the work?

- ☐ a) 40, 60, 120
- ☐ b) 60, 90, 180
- ☐ c) 20, 30, 60
- ☐ d) 30, 45, 90

Solution:

Let R take $6x$ days to complete the work.

$\therefore$ P and Q will take $2x$ and $3x$ days respectively.

$$\frac{1}{6x} + \frac{1}{2x} + \frac{1}{3x} = \frac{1}{10} \text{ (given)}$$

$$\Rightarrow x = 10$$

$\therefore$ P, Q and R take 20 days, 30 days and 60 days respectively to complete the work.
Choice (C)

Directions for questions 6 to 10: Select the correct alternative from the given choices.

P is a 400-digit positive integer. Q is the number formed by reversing the digits of P.

Consider the following statements:

I: The difference of P and Q is divisible by 11.

II: The difference of P and Q is divisible by 9.

Which of the following can be concluded?

- ☐ a) Only I is true
- ☒ b) Only II is true (Correct Answer - ✓)
- ☐ c) Both I and II are true
- ☐ d) Neither I nor II is true

Solution:

The difference of any N – digit positive integer and its reverse is necessarily divisible by 9. It is necessarily divisible by 11 only when N is odd.

The number of digits that P has is even.

∴ The difference of P and Q is divisible by 9 but not necessarily by 11.

Choice (B)

Directions for questions 6 to 10: Select the correct alternative from the given choices.

An amount of Rs.16,700 lent at 15% per annum under simple interest earned Rs.5625 less interest than the interest earned when Rs.24,200 is lent at the same rate for the same period. What was the time period for which the amount was lent?

- ☐ a) 3 years
- ☐ b) 4 years
- ☐ c) 5 years
- ☐ d) 6 years

Solution:

As per the given data, the difference of the sums given earned an interest of ₹5625 in T years (say).

$$\therefore 5625 = \frac{(24200 - 16700) \times 15 \times T}{100}$$

$$\Rightarrow T = 5 \text{ years.}$$

Choice (C)

Directions for questions 6 to 10: Select the correct alternative from the given choices.

Lara, a model, is on a controlled diet and consumes a 600 gm mixture of food P and food Q every day. P contains 10% protein and Q has 22% protein. If she consumes 114 gm protein daily, how many grams of Q is there in the mixture?

- ☐ a) 450 gms
- ☐ b) 150 gms
- ☐ c) 300 gms
- ☐ d) None of these

Solution:

She consumes 114 gm of protein in total

$$\text{i.e. } \frac{114}{600} \times 100 = 19\%$$

∴ By Alligation,

$$\begin{array}{ccc} 10 & & 22 \\ & \searrow \quad \swarrow & \\ & 19 & \\ & \swarrow \quad \searrow & \\ 9 & & 3 \end{array}$$

Therefore, the mixture has P and Q in the ratio 1 : 3.

$$\therefore 9 \text{ parts i.e. weight of Q} = \frac{600}{12} \times 9 = 450 \text{ gms.}$$

Choice (A)

Directions for questions 6 to 10: Select the correct alternative from the given choice

In triangle PQR,

$\angle$

$$P = 120^\circ,$$

$\angle$

$$Q = 15^\circ \text{ and } QR = 2$$

$$(\sqrt{3} - 1)$$

. Find the area of triangle PQR (assume $\sin 15^\circ =$

$$\frac{\sqrt{3} - 1}{2\sqrt{2}}$$

)

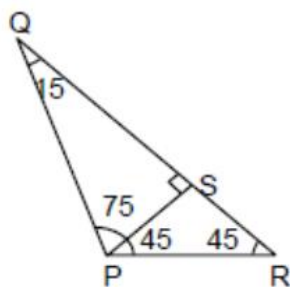
☐ a) $\frac{\sqrt{2}}{3}(6\sqrt{3} - 10)$

☐ b) $\frac{2}{\sqrt{3}}(6\sqrt{3} - 10)$

☐ c) $\frac{\sqrt{2}}{\sqrt{3}}(6\sqrt{3} - 10)$

☐ d) $\frac{6\sqrt{3} - 10}{\sqrt{3}}$

Solution:



Applying sine rule, $\frac{QR}{\sin P} = \frac{PR}{\sin Q} = \frac{PQ}{\sin R} = 2R$

$$\frac{QR}{\sin 120^\circ} = \frac{PR}{\sin 15^\circ} \Rightarrow \frac{QR}{\frac{\sqrt{3}}{2}} = \frac{PR}{\frac{\sqrt{3}-1}{2\sqrt{2}}}$$

$$QR = 2(\sqrt{3}-1)$$

$$\therefore PR = \frac{2(4-2\sqrt{3})}{\sqrt{6}}$$

Area of triangle PQR

$$\begin{aligned} &= \frac{1}{2} (PQ) (PR) \sin 45^\circ = \frac{1}{2} 2(\sqrt{3}-1) \frac{2(4-2\sqrt{3})}{\sqrt{6}} \frac{1}{\sqrt{2}} \\ &= \frac{1}{\sqrt{3}} (6\sqrt{3}-10) \end{aligned}$$

Choice (D)

Directions for questions 11 and 12: Type in your answer in the input box provided in the question.

The average of 21 numbers is 35. None of them is less than 20 and at least 11 of them are not less than 30. The greatest possible value of any of the 21 numbers is .

Your Answer: Unattempted

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Solution:

To get the value of the greatest of the 21 numbers, we have to select the least possible values for the other 20 numbers, subject to the given constraints.

There are at least 11 which are not less than 30. Therefore, we can set 10 numbers equal to 30. The eleventh would be the greatest. None of the numbers is less than 20. Therefore, we set the other 10 number equal to 20.

The negative deviation from the average of 35 is $10(5) + 10(15)$, viz 200. Therefore, the positive deviation from 35 is also 200, i.e., the greatest possible value is $35 + 200$, viz 235.

Ans : (235)

Directions for questions 11 and 12: Type in your answer in the input box provided in the question.

In a class, the number of girls is 2.5 times the number of boys. Initially, each student was asked to submit n essays, where n is one more than the number of boys. Later, each boy was asked to team up with each girl, by turns, and each pair was asked to undertake a small project and submit a report. If the number of essays submitted is 385, find the number of project reports submitted .

Your Answer: Unattempted

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Solution:

Let the number of boys be $2n$

$\therefore$ The number of girls is $5n$ and the number of essays each student had to submit is $2n + 1$

$$\therefore 7n(2n+1) = 385 \Rightarrow n(2n+1) = 55 \Rightarrow n = 5 \quad (\because n > 0)$$

The number of boys and girls is 10 and 25 respectively. The number of project reports submitted is 250.

Ans : (250)

Directions for questions 13 to 15: Select the correct alternative from the given choices.

$$\frac{1}{3^2 - 1}$$

$$\frac{1}{5^2 - 1}$$

$$\frac{1}{21^2 - 1}$$

_____.

☐ a) $\frac{10}{11}$

☐ b) $\frac{10}{21}$

☐ c) $\frac{5}{22}$

☐ d) $\frac{7}{11}$

$$\frac{1}{3^2-1} = \frac{1}{2 \times 4}$$

$$\frac{1}{5^2-1} = \frac{1}{4 \times 6}$$

$$\frac{1}{7^2-1} = \frac{1}{6 \times 8}$$

$$\frac{1}{21^2-1} = \frac{1}{20 \times 22}$$

$$= \frac{1}{3^2-1} + \frac{1}{5^2-1} + \dots$$

$$= \frac{1}{2 \times 4} + \frac{1}{4 \times 6} + \frac{1}{6 \times 8} + \dots + \frac{1}{20 \times 22}$$

$$= \frac{1}{4} \left[\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots + \frac{1}{10 \cdot 11} \right]$$

$$= \frac{1}{4} \left[1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{10} - \frac{1}{11} \right]$$

$$= \frac{1}{4} \cdot \frac{10}{11}$$

$$= \frac{5}{22}$$

Choice (C)

Directions for questions 13 to 15: Select the correct alternative from the given choices.

$$\text{If } a^2 > 1 \text{ and } \left(\frac{a^p}{a^{q+r}}\right)^3 \left(\frac{a^q}{a^{p+r}}\right)^4 \left(\frac{a^r}{a^{p+q}}\right)^5$$

=

$$\left(\frac{a^r}{a^{4q}}\right)^{-2}$$

where p, q, r are integers, the value of p + 6q can be _____.

- ☐ a) 6
- ☐ b) 12
- ☐ c) 26
- ☐ d) 42

Directions for questions 13 to 15: Select the correct alternative from the given choices.

One black ball, two identical green balls, one orange ball and four identical grey balls are arranged in a row such that no two adjacent balls are of the same colour. If the first ball and the last ball are of different colours, in how many ways can they be arranged?

- ☐ a) 12
- ☐ b) 18
- ☐ c) 24
- ☐ d) 36

Solution:

Since there are four grey balls, either they can be at 1st, 3rd, 5th, 7th position or 2nd, 4th, 6th, 8th position.

The remaining balls can be arranged in $\frac{4!}{2!}$ ways

i.e. 12 ways

∴ Total number of ways = $12 \times 2 = 24$ ways.

Choice (C)